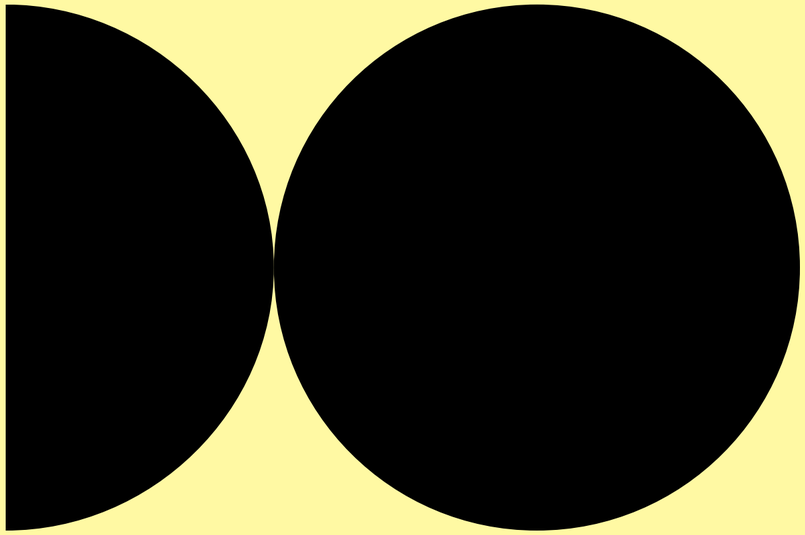
Team Logging off Early
Adnan Nalawala Saifuddin
Arnold Otieno Soi Odhiambo



Role-aware log intelligence on SereneDB

SQL COMPUTES. THE LLM EXPLAINS.

EXHIBIT

THREE READERS

One question, three readers: which problem should we fix first?



Engineer

Diagnose: what broke, on which machine, and what does the line say?
Raw lines · phrase search · timeline



Team lead

Assign: whose work is it, and is it getting worse?
Signatures by owning team · weekly trend · no raw lines



Manager

Prioritise: what first, and why should I believe it?
Risks by blast radius and persistence · no log text

One node in 6h22m (2005-06-11)

152,183

alert lines from ONE node: 44% of all alert lines. Days later a fleet-wide event (2005-06-14) wrote only 49,154 lines across 22,529 nodes.

So we count incidents: 348,460 alert lines → 19,250 incidents (5-min gap) → 1,377 per signature.

LINES ARE NOT INCIDENTS

data TLB error

Ranked by lines: 152,734 lines, but only 117 incidents, and 99.64% of them from one node.

Lustre + ciod

Ranked by incidents: Lustre mount 3,667 and ciod socket 3,638, with at most 8 nodes in any one incident.

rts panic

Ranked by blast radius + persistence: 3,394 nodes over 124 active days. The data TLB error comes 12th.

THREE RANKINGS

Same BGL corpus, three orderings, three different first problems.

HOW EXHIBIT WORKS

01 · Any MCP client
asks as a persona, quotes evidence IDs, never writes SQL

02 · Exhibit MCP server
12 tools, bound parameters, envelope + ledger, answer lints

03 · SereneDB 26.09.1.3
inverted indexes (@@), incident views, roles + GRANTs

The client asks. SereneDB computes. Every answer carries its query and evidence ID: every figure comes from SQL that ran.

TWELVE TOOLS, ONE ENVELOPE

Orient + find

describe_data · ownership · search_logs · nearby_events

Measure

count_events · episodes · top_signatures · trend · what_changed · durations

Trust

verify_evidence · lint_answer

Every result: rows · SQL · params · true total · truncated · evidence ID.

Limits: at most 200 rows per answer · 400 buckets per trend · log text quoted, never dumped · 30-second deadline, enforced client-side.

THREE WAYS TO CONNECT

01

Clone the repo

.mcp.json registers the server
uv run exhibit-mcp (stdio)

02

Local HTTP

make mcp-http
127.0.0.1:8900/mcp

03

Live, zero setup

Streamable HTTP · read-only · same database roles
https://hermes.bushbaby-platy.ts.net:10000/mcp
Tested 18 Oct: 12 tools; E-55EF08 identical to a local run; verify_evidence: MATCH

ROLES ENFORCED BY THE DATABASE

count_events(bgl, group_by=team): one call, three personas.

Not permitted: lead → search_logs(bgl, "ciod"). SereneDB refuses: an index search is privilege-checked against the base table, and the lead role has no grant on it.

Engineer · 4,747,963

rows, counted from the raw lines

Team lead · 4,747,963

lines, summed from a daily rollup

Manager · 348,460

alert lines only, no log text

W1 · Engineer

ONE MACHINE, NOT THE FLEET

152,734

lines from a phrase search for "data TLB error interrupt"

117

incidents at a 5-min gap. The gap is a parameter: 132 at 1 min, 57 at 60 min.

99.64%

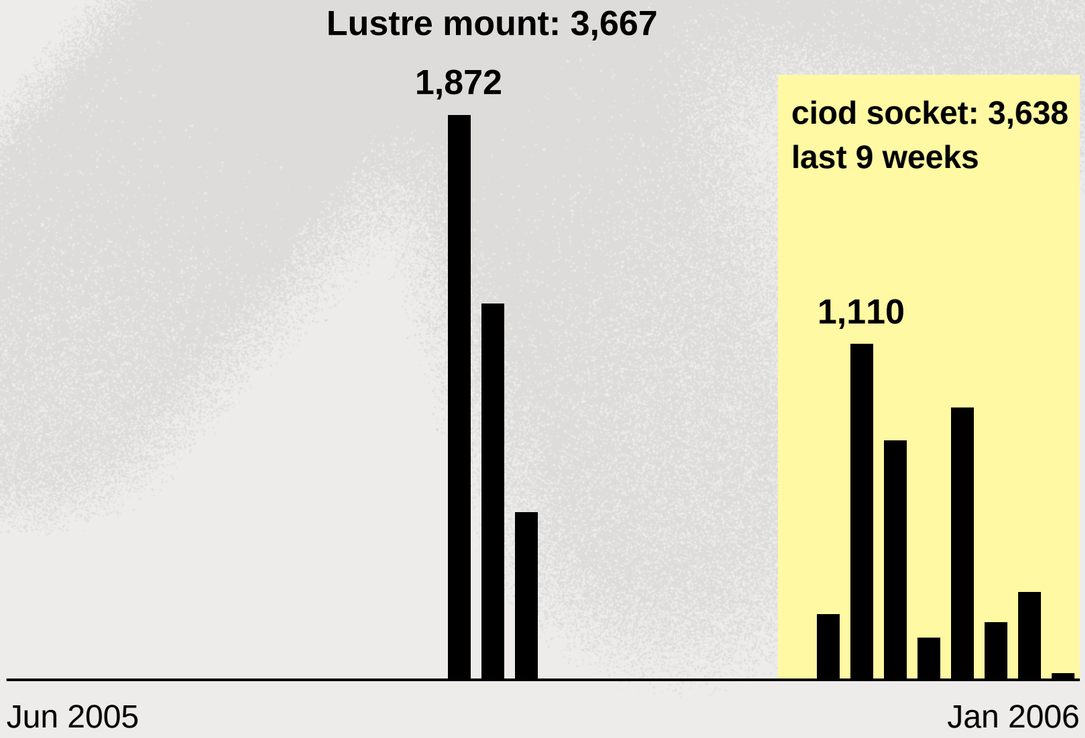
of lines from one node. Deepest incident: R30-M0, 152,182 lines on 1 node in 6:21:47.98 (E-55EF08).

Next step: pull that node's neighbouring events (nearby_events) before touching the fleet.

W1 · Team lead

ONE STOPPED, ONE STILL ACTIVE

Incidents per week, 5-min gap, Jun 2005 to Jan 2006



91.1% of lines, 51.7% of incidents

compute-platform's shares (81.6% of alert lines). workload-runtime holds 48.0% of incidents.

Lustre mount · 3,667 incidents

stopped after 3 weeks; the logs do not say why

ciod socket · 3,638 incidents

every one in the last 9 weeks, and still active

Next step: assign the ciod socket error to workload-runtime now.

W1 · Manager

A SLOW BURN OUTRANKS A FIRE

Not established by the logs: cost, downtime, impact on jobs or users, root cause. The manager's view holds no log text and no node IDs.

19,250

incidents from 348,460 alert lines, ranked by blast radius (distinct nodes) + persistence (active days)

#1

rts panic: fix first. KERNEL · compute-platform · 3,394 nodes · 124 active days

#12

data TLB error: 1st by lines (152,734), but 541 nodes · 11 days · 99.64% on one node

W2 · OpenStack

ONLY LIFECYCLE FINDS THE SLOW ONES

01

Errors: noise

188 ERROR + CRITICAL lines: 8 tracebacks of one image-cache bug, plus 2 token-validation incidents

02

Latency: identical

API latency: 0.2340 s abnormal vs 0.2341 s normal.
The labels sit in anomaly_labels.txt, scored by a script the agent cannot call.

03

Lifecycle: separates

Spawn time ranks the 4 labelled instances 1 to 4 of 2,069:
a445709b 50.72 s · 544fd51c 41.81 s
1643649d 31.27 s · ae651dff 29.46 s
(next 26.69 s; median 19.74 s)

EVERY NUMBER TRAVELS WITH ITS QUERY

```
> episodes(bgl, persona=engineer, gap_minutes=5)
evidence_id: E-55EF08 · role: exhibit_engineer
total_matches: 19,250 · returned: 1 · truncated: true
row: data TLB error interrupt · R30-M0 · 152,182 lines · 1 node · 6:21:47.98

> verify_evidence("E-55EF08")
MATCH: 19,250 rows recorded, 19,250 now
```

Re-runnable

the ledger keeps SQL, role and result hash

Linted

a manager answer that claims downtime or revenue is rejected

Scored

make check 101/101 · make eval 15/15 questions · 19/19 figures, 6 refusals

1 to 4 of 2,069

OpenStack: where the 4 labelled instances rank. By chance: 1 in 761,325,852,001.

99.9998%

BGL agreement on 4,747,963 lines. BGL's label is in the data the agent can read, so here we test agreement, not blindness.

0.999991

precision of signatures (FATAL/FAILURE level: 0.4065). Negative controls: 0 of their 100 lines above INFO. make validate: 13/13.

SCORED AGAINST ANSWER KEYS

make validate runs outside the agent. anomaly_labels.txt is read from disk; no tool, table or column holds it.

```
SET enable_external_file_cache = false;
CREATE VIEW bench_otel_slice AS
  SELECT ... FROM read_parquet(<S3 url>)
  WHERE "TimestampTime" in a 10 s window;
CREATE INDEX bench_otel_idx
  ON bench_otel_slice USING inverted (...);
```

A BILLION ROWS, SEARCHED WHERE THEY SIT

The organisers' OTel file on S3

1B

rows: 41 GB, 1,624 row groups, never downloaded; its shape read from the footer in about 2 s.

A 10 s slice, indexed where it sits: 5,275,597 rows (10 of 1,624 row groups), built in 60 s, 136 MB on disk; phrase + facet search in 13 ms, 290,473 hits.

30 s and 60 s slices crash the build on our machine: reported, not hidden (16 vCPU, about 6 MiB/s to S3).

WHAT WE DO NOT CLAIM

Ownership
The team map is ours: 23 rows, fictional teams, 100% coverage.

The brief vs the data
Brief: 12 categories, 5 levels. Data: 42 and 6. Organisers: the data wins.

Impact
No cost, downtime, customer or revenue figure: the logs cannot support one.

84,253 ASSERT lines are labelled normal; only the message text shows they are faults.

EXHIBIT

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SchubKI Hackathon 2026

RUN IT YOURSELF

Every number in this deck is printed by one of these commands.

```
git clone github.com/Arnie1x/exhibit
make up fetch parse
make bootstrap   # about 20 s
make smoke demo check eval validate
make bench   # the 1B-row file
make mcp   # or the live endpoint
```

SQL COMPUTES.
THE LLM EXPLAINS.